

Mohamed Hasabalah

SOFTWARE ENGINEER - .NET

📍 New Cairo, Egypt 📞 +20 109 974 6971 ✉ mohamedhasabalaah@gmail.com [in mohamed-hasabalah](#) [mhasabalah](#)

PROFESSIONAL SUMMARY

.NET engineer with **6+ years** architecting enterprise ERP, logistics and real-estate platforms across Egypt and Saudi Arabia, currently decomposing a multi-team ERP monolith into independently deployable modules. Builds event-driven systems that hold under at-least-once delivery: schema-per-module boundaries, RabbitMQ and MassTransit messaging, transactional outbox, and idempotent consumers. Owns domains end to end, from DDD modelling and API design through Azure deployment, CI/CD, observability and production hardening.

TECHNICAL SKILLS

LANGUAGES & FRAMEWORKS C#, .NET 9/10, ASP.NET Core, Minimal APIs, REST APIs, Entity Framework Core, .NET Aspire, gRPC, SQL, Blazor, Vue.js, TypeScript

ARCHITECTURE Modular Monolith, Vertical Slice Architecture, Domain-Driven Design, CQRS (MediatR, FluentValidation), Event Sourcing, Clean Architecture, Event-Driven Architecture, Microservices

DISTRIBUTED SYSTEMS RabbitMQ, MassTransit, Azure Service Bus, Hangfire, Transactional Outbox, Inbox Pattern, Idempotent Consumers, Optimistic Concurrency, Saga Orchestration, Dead-Letter Queues, Retry with Backoff and Jitter, Redis

CLOUD & DEVOPS Azure (App Service, Functions, Service Bus, Key Vault, Application Insights), Bicep IaC, Docker, Docker Compose, GitHub Actions, GitHub Container Registry, Azure DevOps, CI/CD, Zero-Downtime Deployment

DATA PostgreSQL, SQL Server, Oracle, MongoDB, EventStoreDB, Redis, Query Tuning, Execution Plan Analysis, Indexing, Replication

OBSERVABILITY & TESTING OpenTelemetry, Prometheus, Grafana, Loki, Serilog, Seq, Application Insights, xUnit, NSubstitute, Testcontainers, Integration & E2E Testing

SECURITY Keycloak, OAuth2, OpenID Connect, JWT, Secrets Management

PRACTICES Agile, Scrum, Kanban, Code Review, Technical Mentoring

EXPERIENCE

MicrotecSaudi - Software Engineer

Nov 2024 - Present / Cairo

- Re-architected **6 core ERP modules** from a shared-database monolith into independently deployable services with schema-per-module ownership and public contracts, eliminating cross-schema coupling.
- Replaced synchronous calls with an event-driven backbone on RabbitMQ, MassTransit and Azure Service Bus, using competing consumers, jittered backoff and dead-letter queues to prevent cascades.
- Moved **2 workloads**, bulk imports and notification dispatch, onto **Hangfire** background jobs with retries and scheduled runs, so failures recover without blocking users.
- Operated the platform on **4 Azure services** (App Service, Functions, Key Vault, Application Insights), with environments provisioned in **Bicep** and containers on Docker Compose.
- Cut new-report delivery **from a release cycle to same-day configuration** with a reporting service driven by runtime definitions, absorbing 15+ report requests a month.
- Owned Accounting and HR modules end to end, alongside the Blazor Admin Portal.
- Standardised UI across **6+ product teams** with a reusable Blazor component and services package published as internal NuGet.
- Proved each module extraction safe before release with **unit tests** on domain logic and **integration tests** across boundaries.
- Set the review standard for module boundaries and messaging patterns, and mentored **6 engineers** onboarding onto the distributed codebase.

MSDC - Software Engineer

Sep 2023 - Oct 2024 / Cairo

- Modelled ERP domains with Domain-Driven Design using bounded contexts and aggregates that enforce their own invariants, then decomposed them into services over RabbitMQ.
- Cut month-end reporting **from roughly 45 minutes to under 10** by rewriting the heaviest Oracle queries, correcting indexing and eliminating full scans.
- Automated document generation with QuestPDF across **10 document types**, eliminating roughly 6 hours per week of manual report preparation.

- Built a logistics platform across five modules covering policies, shipments, invoices and real-time tracking, processing **1,000+ invoices and 200+ shipments daily**.
- Took releases **from monthly and manual to several per week** with CI/CD on GitHub Actions, removing a recurring source of production errors.
- Untangled a legacy codebase into clear service boundaries and made them part of the code-review standard, lowering the cost of every feature that followed.

INNOTECH - Software Engineer

Mar 2020 - May 2022 / Tanta

- Built ASP.NET Core services and Blazor WebAssembly interfaces, modelling the SQL Server schema behind **25+ application features**.
- Established code review as a team practice and standardised C# conventions across the codebase, raising consistency as the team grew.
- Owned features end to end, from data model and API through to UI, testing and deployment.

PROJECTS**BateCom** - Real-estate platform, .NET 9

Product, delivery & operations

- Built a **real-estate marketplace** across **6 modules** (Properties, Developers, Agents, Subscriptions, Attachment, Identity) where developers and agents publish listings behind subscription billing.
- Offloaded listing imports and notification dispatch to **Hangfire** background jobs, keeping long-running work off the request path.
- Consolidated local development onto a **.NET Aspire** AppHost orchestrating **4 resources**: API, PostgreSQL, Redis and Seq.
- Mirrored it in production with a **3-file** Docker Compose split into core, observability and dev-only tooling, so environments never drift.
- Automated delivery with **7 GitHub Actions workflows** on a self-hosted runner: tests, image publishing to GHCR, scheduled database backups, health monitoring and one-click rollback.
- Scripted **zero-downtime releases** with pre-deploy backup, deploy and rollback behind **nginx** TLS termination, with health checks reverting a bad release automatically.
- Hardened production with read-only Docker secrets, localhost-bound service ports behind nginx, and an internal-only database network.
- Instrumented with **OpenTelemetry** into **3 backends** (Prometheus, Tempo, Loki), surfaced in Grafana alongside Serilog logging in Seq.

Flight Booking Platform - .NET 10

Data consistency & concurrency

- Built an airline reservation system across **4 modules** (Flight, Passenger, Booking, Identity), each owning its data and communicating over typed gRPC contracts.
- Separated read and write sides across **2 stores**, PostgreSQL for writes and MongoDB for reads, kept current by integration events.
- Persisted the Booking aggregate to **EventStoreDB** as an append-only stream, giving a full audit trail and rebuildable projections.
- Prevented **double-booked seats** using EF Core optimistic concurrency on a row version, returning 409 Conflict on stale writes with an explicit client retry contract.
- Verified with integration tests running **3 real containers** (PostgreSQL, MongoDB, EventStoreDB) through Testcontainers.

Modular E-Commerce - .NET 8

Messaging reliability & access control

- Implemented the **transactional outbox** in **2 modules**, Basket and Ordering, so a state change and its integration event commit atomically under at-least-once delivery.
- Made consumers **idempotent** through message deduplication, so a broker redelivery cannot place a duplicate order or double-apply a basket change.
- Secured the platform with Keycloak (OAuth2, OpenID Connect) and cut repeat catalogue reads with Redis caching on the hottest query paths.

EDUCATION**Bachelor of Engineering - Computer Engineering and Automatic Control**

2018 - 2023

Tanta University, Tanta, Egypt - Graduated with Very Good honours

LANGUAGES

Arabic - Native / English - Professional working proficiency